Case 05: Firebrand Transport in Time-Dependent Wind

1 Verification purpose

This case verifies Eulerian firebrand transport when wind varies in space and time. Surface spread, crosswind dispersion, consumption, and ignition delay are disabled so the leading edge follows a controlled characteristic equation.

The test exposes incorrect meteorology interpolation, direction conversion, tracker lifetime, or transport/level-set timestep coupling. Success requires leading-edge position and running-average spread rate to agree with an independently integrated characteristic solution.

2 Mathematical formulation and numerical implementation

The imposed one-dimensional velocity is

$$u(x,t) = \bar{u} + A \sin\left(\frac{2\pi x}{\lambda_x}\right) \sin\left(\frac{2\pi t}{T}\right), \quad (\bar{u}, A, \lambda_x, T) = (6.71, 3.355, 100, 60) \tag{1}$$

in SI units [1]. The exact leading edge is the characteristic satisfying

$$\frac{dx_{\mathrm{LE}}}{dt} = u(x_{\mathrm{LE}}, t), \qquad x_{\mathrm{LE}}(0) = 5 \text{ m}. \tag{2}$$

The postprocessor integrates this ODE with fourth-order Runge–Kutta and a 0.005 s substep.

The preprocessor samples the analytical wind every 1 s into 121 bands of wind speed; wind direction holds a constant 270-degree meteorological direction. ELMFIRE linearly interpolates adjacent weather bands using meteorological sampling interval. The namelist explicitly selects area-proportional generation, Eulerian accumulation, and immediate ignition. Because surface fire is disabled, the per-area source avoids any dependence on fireline intensity. Each ignited cell emits for 10 s.

3 Derivation of expected behavior

The characteristic velocity is strictly positive: $3.355 \le u \le 10.065$ m s^{-1}, so the reference leading edge is monotone. At each reporting time the simulated position is reconstructed as the furthest centre-row cell whose final ELMFIRE arrival time is no later than that time. Grid crossing makes the discrete instantaneous rate of spread (ROS) alternate between zero and approximately $\Delta x/\Delta t$. For the reference-aligned diagnostic, a centred nine-point average should recover the smooth reference $u(x_{\mathrm{LE}}(t), t)$.

The maximum prescribed speed used to select the generated solver step is $u_{\max} = 6.71 + 3.355 = 10.065$ m s^{-1}, so

$$\Delta t = 0.5(10)/10.065 = 0.496771 \text{ s}.$$

Independent RK4 integration with the stated 0.005-s substep and $x(0) = 5$ m gives

t (s)	0	30	60	90	120
x_{ref} (m)	5.000	192.328	377.023	566.967	752.389

from the generated analytical wind, not from ELMFIRE output. Although $\sin(2\pi t/60) = 0$ makes the instantaneous reference velocity 6.71 m s^{-1} at these particular reporting times, the integrated position retains the intervening space–time oscillations; for example the 120-s displacement is $752.389 - 5 = 747.389$ m, not a constant-wind substitution.

For matched sample i, the explicit references are $e_{x,i} = x_{\mathrm{ELM},i} - x_{\mathrm{ref},i} = 0$ and $e_{u,i} = \overline{ROS}_{\mathrm{ELM},i} - u(x_{\mathrm{ref},i}, t_i) = 0$. Hence the ideal metric values are mean $|e_x| = 0$, max $|e_x| = 0$, mean $|e_u| = 0$, and coverage $= 1$; the substituted allowances are 10 m, 20 m, 2 m s^{-1}, and 0.90, respectively.

4 Verification metrics and acceptance criteria

For all finite samples, the postprocessor calculates mean and maximum absolute position error, the mean absolute error of the nine-point running-average ROS, and temporal coverage. The case passes only if

$$\mathrm{MAE}_x \leq 10\ \mathrm{m}, \quad \max|e_x| \leq 20\ \mathrm{m}, \quad \mathrm{MAE}_{\overline{\mathrm{ROS}}} \leq 2\ \mathrm{m\,s^{-1}}, \quad f_{\mathrm{coverage}} \geq 0.90.$$

Missing or unusable output is explicitly non-passing.

4.1 Justification of metric quantification

The 10 m position-MAE limit equals one grid cell, and the 20 m maximum limit equals two cells. These bounds recognize that a continuous characteristic is reconstructed from cell-centred arrival events while still rejecting persistent phase errors. Differentiation magnifies that position quantization, so ROS is assessed after a centred nine-point average; its 2 m s^{-1} allowance is smaller than the configured 3.355 m s^{-1} velocity amplitude. A run that merely reproduces the mean wind but misses the oscillatory response therefore cannot satisfy the ROS criterion. The 90% coverage requirement allows a small number of unmatched samples near raster or temporal boundaries but prevents selective omission of a substantial high-error portion of either wind cycle. The four conditions measure bias, worst local error, derivative fidelity, and sampling support, respectively, so all are required for acceptance.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Final time of arrival (TOA) raster with valid physical centre-row coverage; two-cell halo and nodata are excluded.	Usable output required
Position MAE	Mean $\lvert x_{LE,ELMFIRE}(t) - x_{ODE}(t)\rvert$ over finite matched samples.	≤ 10 m
Maximum position error	Maximum absolute leading-edge difference over the same samples.	≤ 20 m
Running-ROS MAE	Mean absolute error of the centred nine-point ELMFIRE ROS against the characteristic solution.	≤ 2 m s^{-1}
Temporal coverage	Fraction of required characteristic times having finite ELMFIRE leading-edge observations.	≥ 0.90
Overall decision	Conjunction of completeness and all four quantitative components.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the base time-dependent-wind run. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

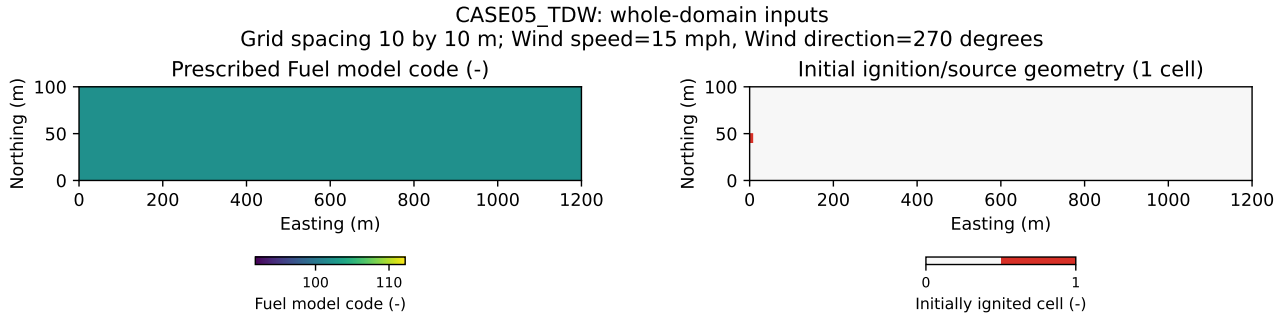


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Table 2: Configuration of the reconstructed time-dependent-wind case.

Parameter	Value	Role
Physical domain	1200 m by 100 m	contains the 120-s trajectory
Cell size	10 m	streamwise discretization
Numerical halo	2 cells per side	excluded from observations
Duration	120 s	two wind periods
Wind bands	121 at 1 s	time-varying forcing
Wind CFL	0.5	resolves wind transport
Generation	PER-AREA, 1 pcs m^{-2} s^{-1}	no-surface source
Emission duration	10 s	fixed finite-duration source
Source selection	100% deterministic	removes sampling variability
Accumulation/ignition	EULERIAN/DIRECT	tested transport and ignition path

The predicted position is a smooth monotone curve. Instantaneous numerical ROS is grid-quantized, while its nine-point mean should follow the oscillatory ODE velocity.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from the base time-dependent-wind run. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The field provides the spatial counterpart to the leading-edge history and reveals whether the changing wind produces a coherent transported front across the complete domain.

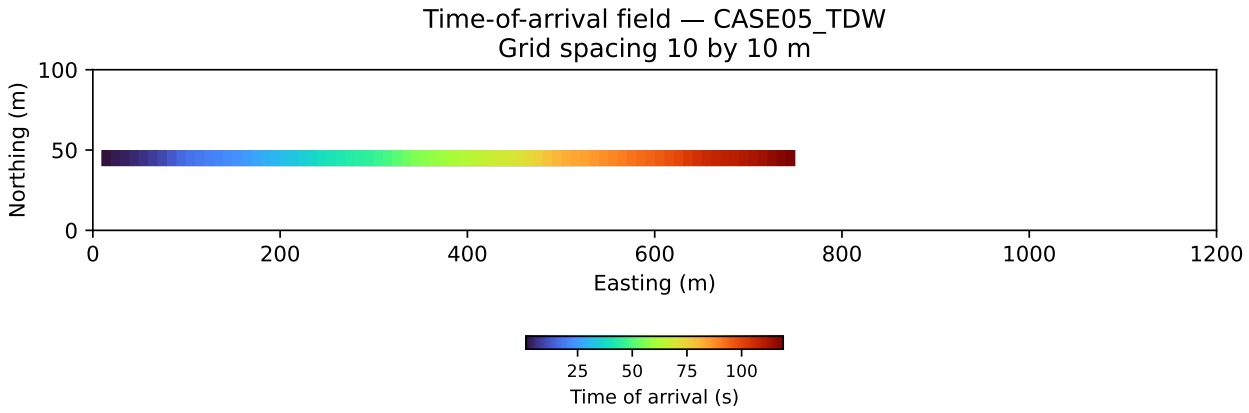


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figure 3 first reproduces the two principal time-dependent-wind observables. The numerical curves are reconstructed from the final ELMFIRE arrival-time raster; the smooth curves are obtained by integrating the prescribed leading-edge ODE.

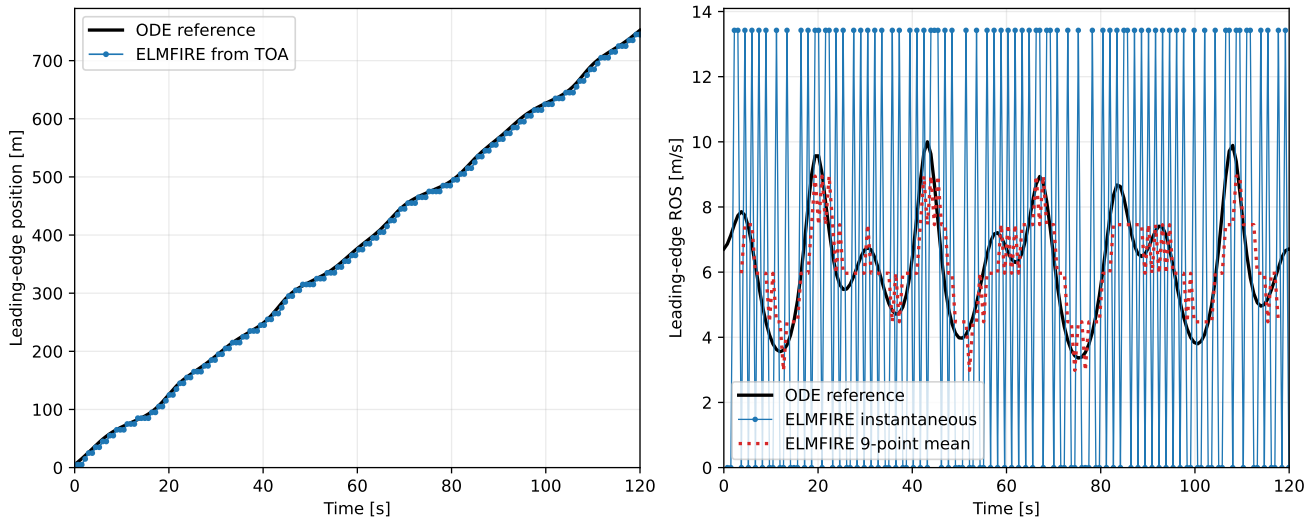


Figure 3: Time-dependent-wind transport comparison. Left: ELMFIRE and ODE leading-edge position. Right: ODE reference, grid-scale instantaneous ROS, and the nine-point running-average ELMFIRE ROS.

Figure 4 adds the time-resolved residuals and acceptance limits. It exposes localized phase errors that can be hidden by a single mean value.

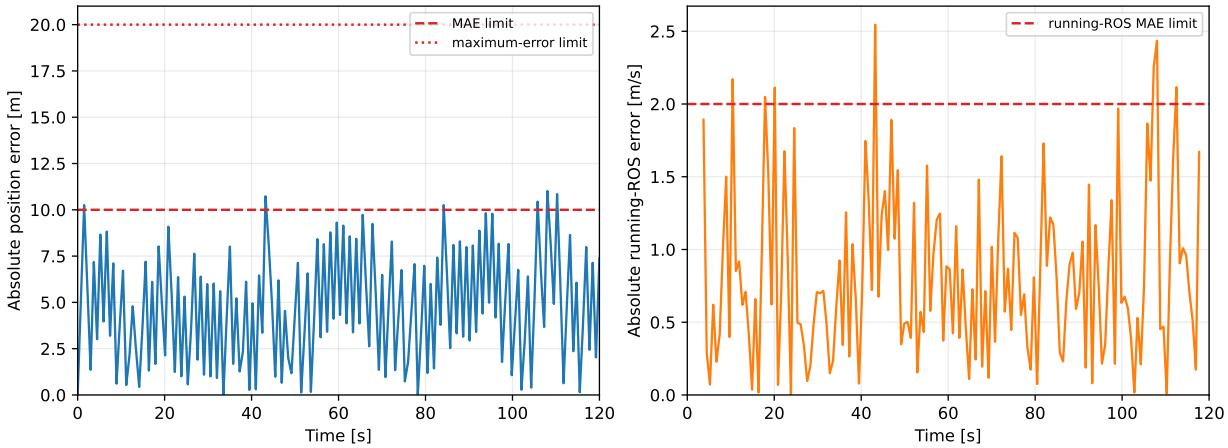


Figure 4: Absolute leading-edge-position and running-ROS errors through the simulation. Horizontal lines identify the applicable acceptance limits.

7 Calculated metrics and verification decision

Table 3: Acceptance metrics calculated by postprocessing.

Metric	Limit	Calculated	Pass
Position MAE	$\leq$10 m	4.761474516898714	True
Position maximum error	$\leq$20 m	11.01743438309552	True
Running ROS MAE	$\leq$2 m/s	0.8128946040373389	True
Time coverage	$\geq$ 0.90	1.0	True

The decision read directly from calculated verification and validation metrics is **PASS** (overall acceptance = True).

8 Technical interpretation

The current execution passes all four predeclared checks. The ELMFIRE leading edge reaches 745.0 m at 120 s, compared with the ODE reference value 752.3894781520776 m. The position MAE is 4.761474516898714 m, the maximum position error is 11.01743438309552 m, the nine-point running-ROS MAE is 0.8128946040373389 m/s, and the time-coverage fraction is 1.0; each lies within its declared limit. Input inspection confirms 121 one-second wind bands spanning the prescribed forcing history and an explicit deterministic source-selection percentage. The agreement therefore verifies the configured time-indexed-wind transport observable at the stated spatial and temporal resolution. Instantaneous ROS remains grid-quantized, which is why the smoother nine-point mean is the acceptance observable.

9 Scope and limitations

This is deterministic numerical verification of the Eulerian algorithm. It does not validate the empirical landing-distance distribution, crosswind dispersion, Lagrangian sampling, or physical ignition proba-

bility. One-second weather bands approximate the continuous analytical forcing by ELMFIRE's native linear interpolation.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.